

Dynamic Hand Gesture Recognition Using LSTM and MediaPipe

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01

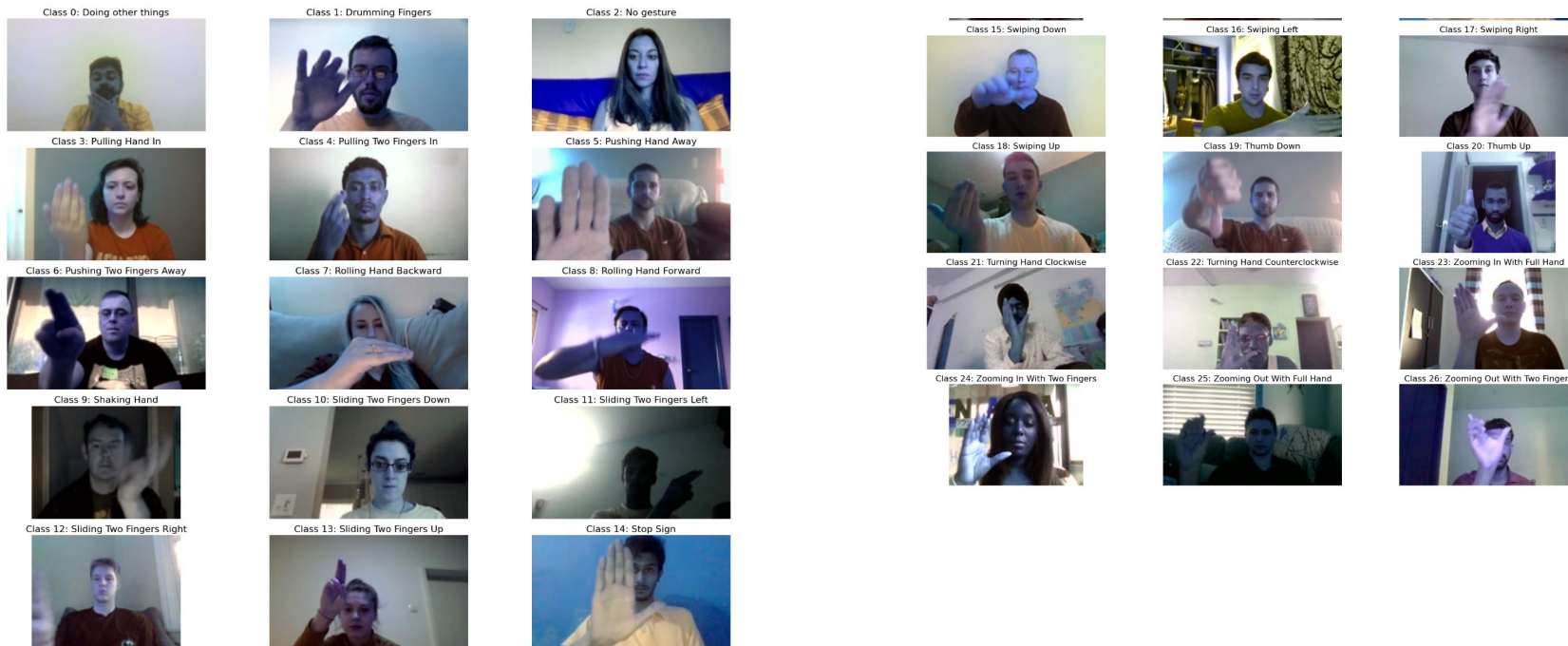
Problem Description

- Dynamic hand gesture recognition is the process of recognizing human hand movements that change over time by analyzing a sequence of frames to understand the motion pattern.
- Examples: 🖐️ Waving, ➡️ Swiping left/right, etc
- More challenging than static hand gesture recognition.

20BN-Jester Dataset:

- **Total Videos:** 64,448 gesture videos
- **Classes:** 27 unique gesture types -> Only 6 classes are used due to resource constraint
- **Training Set:** 118,562 videos -> ~13K Used due to resource constraint
- **Validation Set:** 7047 -> ~2K Used due to resource constraint
- **Test Set:** 6981 videos
- **Resolution:** 100x176 pixels, ~3 seconds each
- **Selected Gesture Classes (6):** Swipe Two Fingers Left, Swipe Two Fingers Right, Swipe Two Fingers Up, Swipe Two Fingers Down, Stop Sign, Doing Other Things.

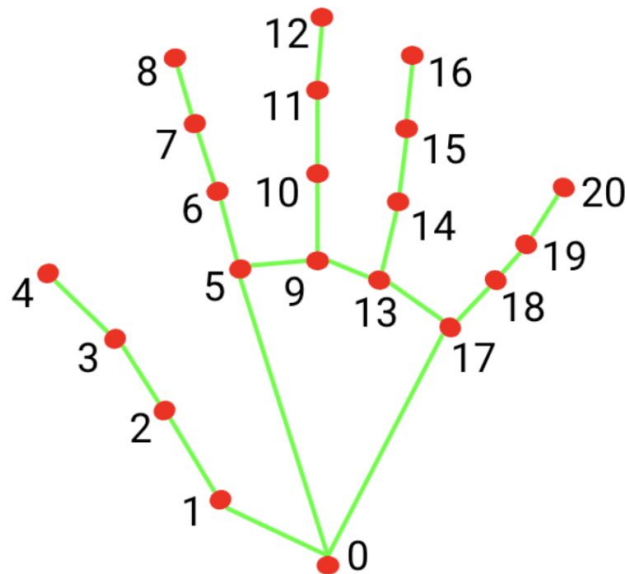
Dataset Overview: 27 Gesture Classes



03 MediaPipe Hand Landmarker Overview

- The MediaPipe Hand Landmarker task lets you detect the landmarks(key points) of the hands in an image.
- This task operates on image data and outputs hand landmarks in image coordinates (x,y,z), hand landmarks in world coordinates and handedness(left/right hand) of multiple detected hands.

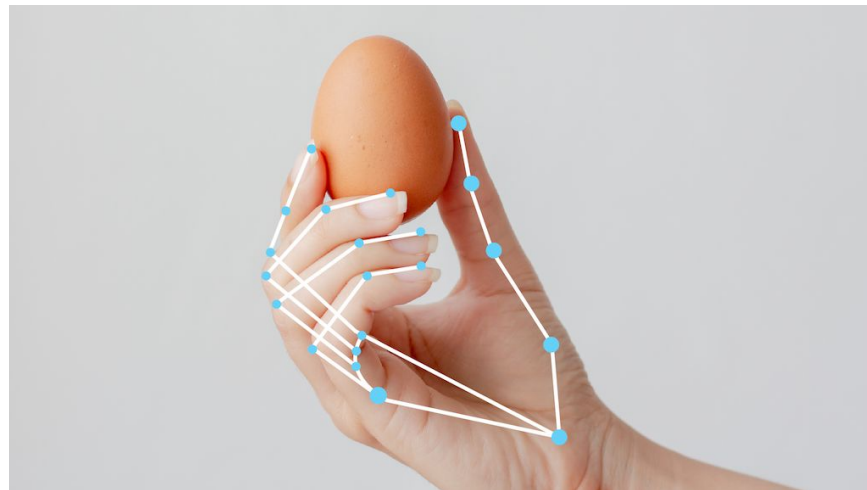
03 MediaPipe Hand Landmarker Overview



- | | |
|-----------------------|-----------------------|
| 0. WRIST | 11. MIDDLE_FINGER_DIP |
| 1. THUMB_CMC | 12. MIDDLE_FINGER_TIP |
| 2. THUMB_MCP | 13. RING_FINGER_MCP |
| 3. THUMB_IP | 14. RING_FINGER_PIP |
| 4. THUMB_TIP | 15. RING_FINGER_DIP |
| 5. INDEX_FINGER_MCP | 16. RING_FINGER_TIP |
| 6. INDEX_FINGER_PIP | 17. PINKY_MCP |
| 7. INDEX_FINGER_DIP | 18. PINKY_PIP |
| 8. INDEX_FINGER_TIP | 19. PINKY_DIP |
| 9. MIDDLE_FINGER_MCP | 20. PINKY_TIP |
| 10. MIDDLE_FINGER_PIP | |

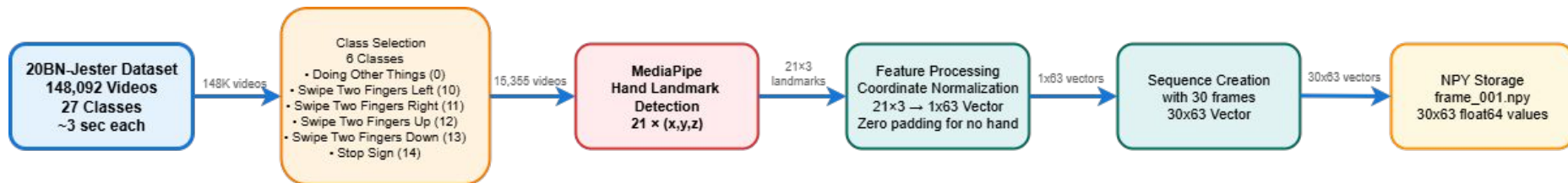
04 Why Use Hand Landmarks ?

- Low-dimensional
- Are not sensitive to lighting/background
- Scale/rotation invariant
- Robust & fast



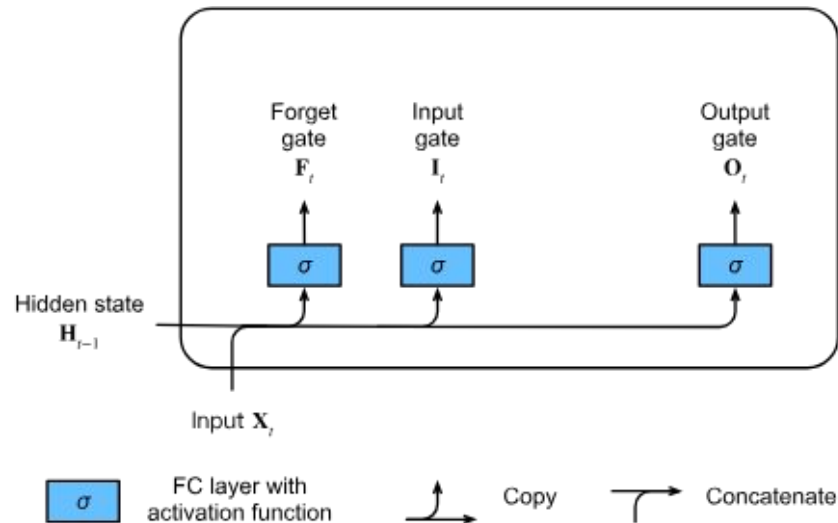
05

Data Preprocessing Pipeline

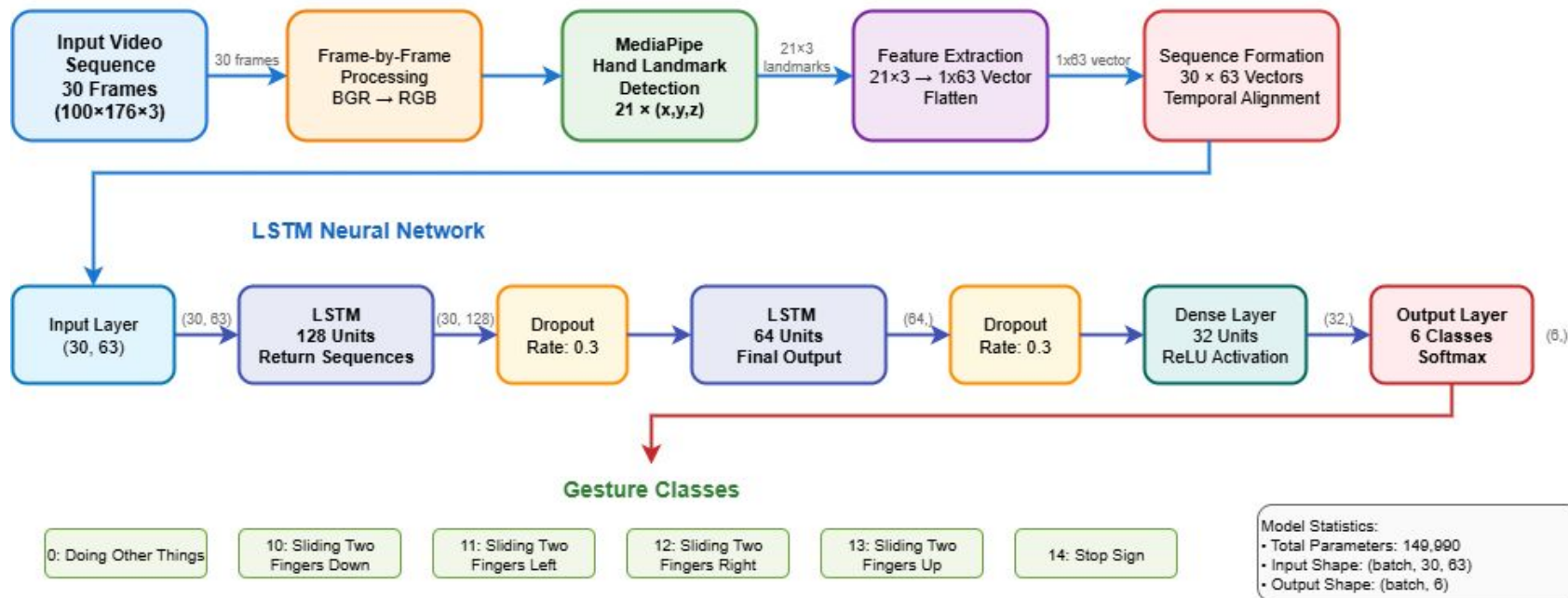


06 Long Short-Term Memory (LSTM) Networks

- Long Short-Term Memory (LSTM) networks are a special kind of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem, which traditional RNNs often encounter when dealing with long sequences.
- Forget Gate** - Decides what to forget
- Input Gate** - Decides what to remember/add
- Output Gate** - Decides what to output

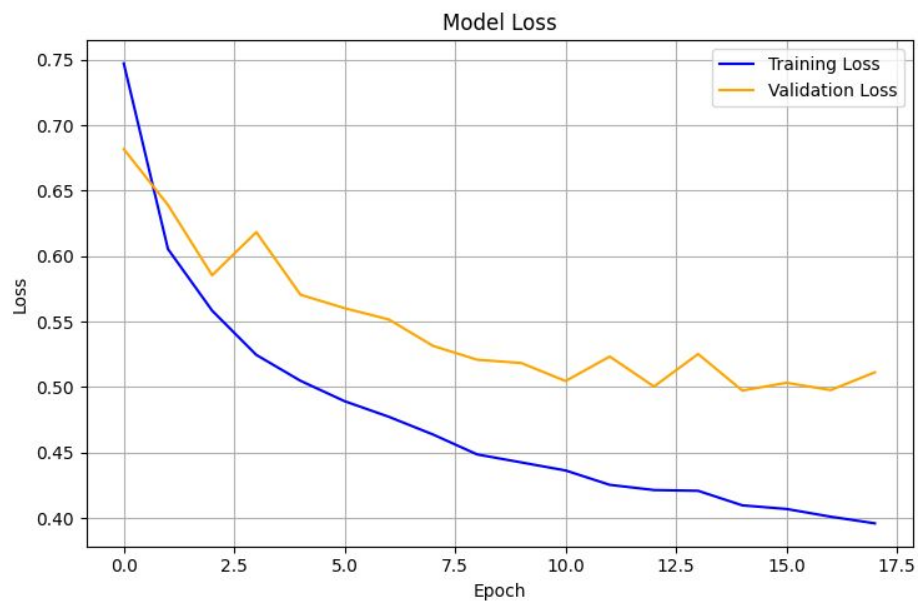
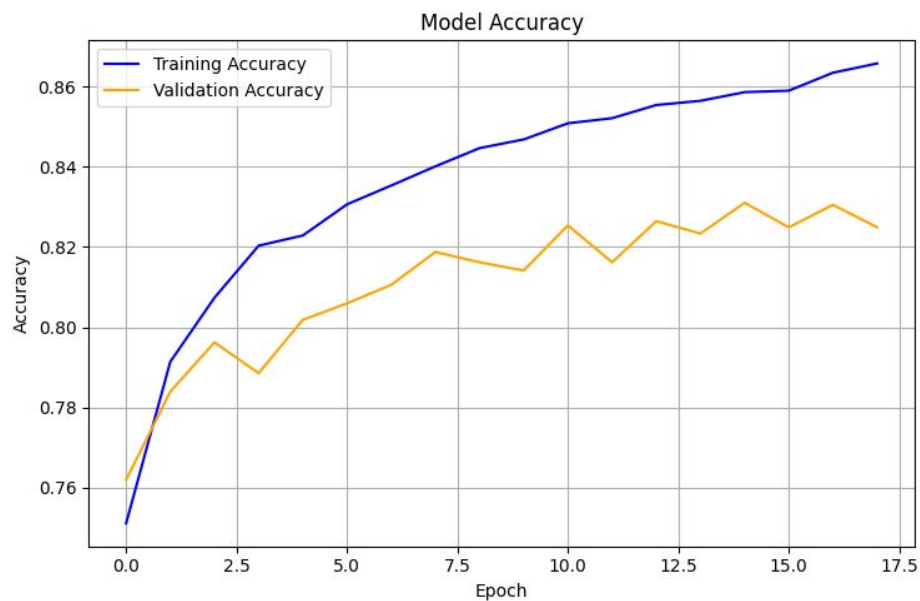


Dynamic Hand Gesture Recognition Architecture



08

Model Evaluation



08 Model Evaluation: Classification Report

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CLASSIFICATION REPORT
=====
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	precision	recall	f1-score	support
Doing Other Things	0.78	0.93	0.85	713
Sliding Two Fingers Down	0.81	0.84	0.82	249
Sliding Two Fingers Left	0.84	0.71	0.77	246
Sliding Two Fingers Right	0.86	0.69	0.77	242
Sliding Two Fingers Up	0.93	0.68	0.79	244
Stop Sign	0.91	0.93	0.92	259
accuracy			0.83	1953
macro avg	0.85	0.80	0.82	1953
weighted avg	0.84	0.83	0.83	1953

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Demo

MediaPipe detector loaded!

Configuration

Confidence Threshold: 0.50

Frame Buffer Size: 30

Process Every N Frames: 1

Choose App Mode: Real-time Video

Instructions

1. Choose Camera Input mode for best compatibility
2. Allow camera access when prompted
3. Position your hand clearly

Live Camera Feed

Gesture: Stop Sign
Confidence: 0.977
Buffer: 30/30

Prediction Results

Real-time predictions will appear here when using video mode

Gesture Classes

- 0: No Gesture
- 10: Slide Down
- 11: Slide Left
- 12: Slide Right
- 13: Slide Up
- 14: Stop Sign

STOP

Manage app

MediaPipe detector loaded!

Configuration

Confidence Threshold: 0.50

Frame Buffer Size: 30

Process Every N Frames: 1

Choose App Mode: Real-time Video

Instructions

1. Choose Camera Input mode for best compatibility
2. Allow camera access when prompted
3. Position your hand clearly

Live Camera Feed

Gesture: Slide Right
Confidence: 0.991
Buffer: 30/30

Prediction Results

Real-time predictions will appear here when using video mode

Gesture Classes

- 0: No Gesture
- 10: Slide Down
- 11: Slide Left
- 12: Slide Right
- 13: Slide Up
- 14: Stop Sign

STOP

Manage app